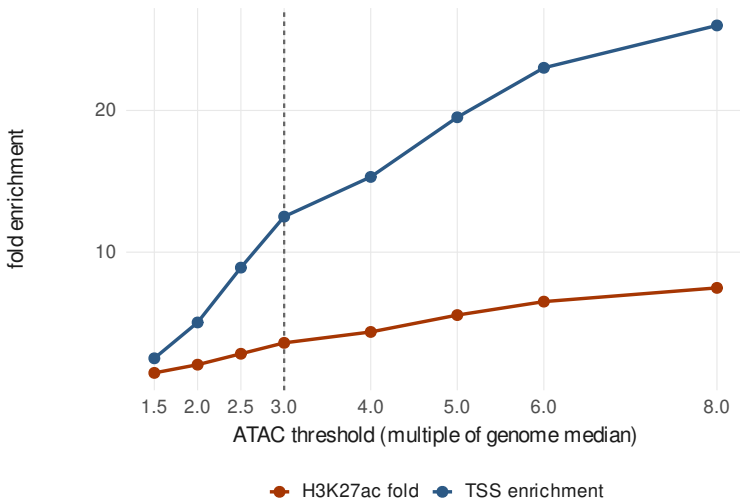


Constructing the paralogue-resolved regulatory layer, and the checks on each step

Supplementary methods. hg38 deposits were lifted to hg19 with 0.41–1.33% loss across 6 files; the ENCODE v2 blacklist removed 5,228 of 102,334 universe regions (5.1%). Panels A and C show guards added after an initial rule failed.

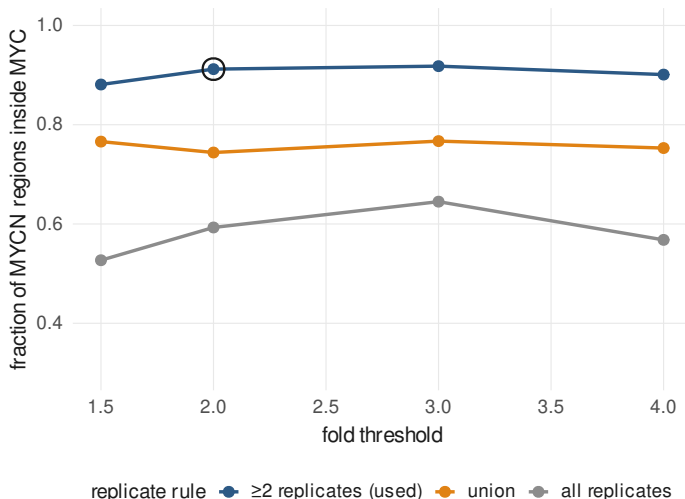
A · Regions carry real regulatory signal

both marks rise monotonically; dashed line is the chosen 3x cutoff



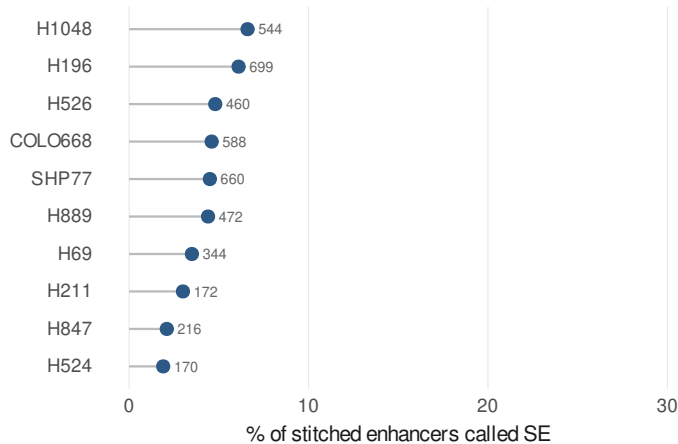
B · Criterion 1 is stable within its convention

MYCN-in-MYC nesting; circled point is the reported value
≥2-replicate rule varies by only 0.037 across thresholds



C · Super-enhancer calls stay far below the guard

% of stitched enhancers called SE; hard guard is 30%
labels give the SE count per line



D · Links are activity-correlated, not nearest-gene

only 32.6% of links agree with the nearest gene;
6.1x enrichment over a distance-matched null

